Complex Contour Integrals and the Geometry of Integration

N-20220813

§1.1 Why complex integration is different

The note begins by comparing real functions with complex functions. If a complex function is regarded merely as a map

$$\mathbb{R}^2 \to \mathbb{R}^2,$$

then it is just a two-dimensional real function. But the complex plane has more structure: it is a field. We can multiply and divide complex numbers, and this extra structure makes holomorphic functions far more rigid than ordinary differentiable maps.

The printed text on the first page highlights two key facts:

(i) contour integrals around closed loops often vanish;

(ii) holomorphic functions are essentially determined by their Taylor series.

The handwritten remark correctly notices that this explains earlier surprising facts: knowing a holomorphic function on a boundary, or on a sequence with a limit point, can determine it inside.

§1.2 Real integrals as a warm-up

For a complex-valued function of one real variable,

$$f(t) = u(t) + iv(t),$$

we define

$$\int_a^b f(t)\,dt = \int_a^b u(t)\,dt + i\int_a^b v(t)\,dt.$$

This is just componentwise integration. Nothing mysterious has happened yet.

The conceptual jump comes when the input is not just a real interval but a path in the complex plane.

§1.3 Contours and parametrization

A contour is a path

$$\alpha : [a, b] \to \mathbb{C}.$$

If f is complex-valued, its contour integral along α is

$$\int_\alpha f(z)\,dz = \int_a^b f(\alpha(t))\alpha'(t)\,dt.$$

This formula is exactly the substitution rule. The factor $\alpha'(t)$ records both speed and direction along the curve.

For the unit circle, the following parametrizations describe the same geometric path run once counterclockwise:

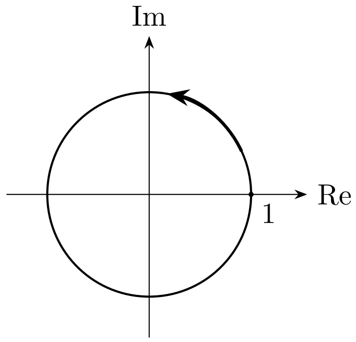
$$\alpha_1(t) = e^{it}, \quad 0 \le t \le 2\pi,$$

$$\alpha_2(t) = e^{2\pi i t}, \quad 0 \le t \le 1.$$

They give the same contour integral. A path such as

$$\alpha_3(t) = e^{2\pi i t^5}$$

also traces the circle once, but at a different speed. The integral is unchanged because the parametrization is merely a way of walking along the same curve.



The unit circle as a contour.

Figure 1.1: A contour integral depends on the oriented geometric path, not on the accidental speed of parametrization.

§1.4 Riemann sums and midpoint sums

The note inserts a real-calculus reminder about Riemann sums. Divide an interval into small pieces Δ_i, choose sample points x_i, and form

$$R = \sum_{i=1}^{n} f(x_i)\Delta_i.$$

As the mesh tends to zero, this approximates the area under the curve.

The trapezoidal rule and midpoint rule are refinements of this same idea. The note's diagrams compare rectangles, trapezoids, and midpoint rectangles. The moral is that integration is not a formula first; it is a limiting process of adding many tiny contributions.

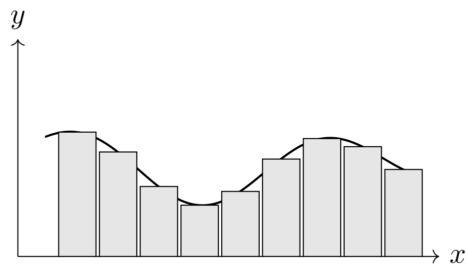


Figure 1.2: A schematic midpoint Riemann sum. The exact picture is less important than the limiting idea.

§1.5 Complex Riemann sums

For a complex path K, divide the path into small directed pieces Δ_i. Choose sample points z_i on the path. The complex Riemann sum is

$$\sum_i f(z_i)\Delta_i.$$

Each term means: take the tiny tangent displacement Δ_i, multiply it by the complex number $f(z_i)$, thereby scaling and rotating it, and then add all these transformed displacements.

This gives the geometric meaning of

$$\int_K f(z)\,dz.$$

A complex integral is not area in the usual sense. It is a limit of transformed tangent vectors.

§1.6 Integrals of powers on the unit circle

The note states the key example:

$$\gamma(t) = e^{it}, \qquad 0 \le t \le 2\pi.$$

Then

$$\int_\gamma z^m\,dz = \int_0^{2\pi} e^{imt} i e^{it}\,dt = i\int_0^{2\pi} e^{i(m+1)t}\,dt.$$

If $m \neq -1$, then the exponential completes full oscillations and cancels out:

$$\int_\gamma z^m\,dz = 0.$$

If $m = -1$, then the integrand is simply i, so

$$\int_\gamma \frac{1}{z}\,dz = 2\pi i.$$

Thus

$$\int_\gamma z^m\,dz = \begin{cases} 2\pi i, & m = -1, \\ 0, & m \neq -1. \end{cases}$$

This is the seed of residue theory. The function $1/z$ is special because it detects the winding of the loop around the origin.

§1.7 Cauchy–Goursat theorem

Theorem 1.7.1 (Cauchy–Goursat theorem)

If f is holomorphic on a simply connected domain D, then for every closed contour $\gamma \subset D$,

$$\int_\gamma f(z)\,dz = 0.$$

The last page states the Cauchy–Goursat theorem: if f is holomorphic on a simply connected domain Ω, then for every closed contour γ in Ω,

$$\int_\gamma f(z)\,dz = 0.$$

The note compares two paths with the same endpoints and suggests path independence in simply connected domains. This is the right intuition. If every loop can contract without crossing a singularity, holomorphic integrals do not accumulate circulation.

§1.8 Contour integrals depend on paths

For a complex function, the integral

$$\int_\gamma f(z)\,dz$$

depends on both the function and the path γ. A parametrization $z = \gamma(t)$ gives

$$\int_\gamma f(z)\,dz = \int_a^b f(\gamma(t))\gamma'(t)\,dt.$$

Thus complex integration is line integration in the plane with complex multiplication built in.

§1.9 Primitive functions and path independence

If $f = F'$ on a domain, then

$$\int_\gamma f(z)\,dz = F(\gamma(b)) - F(\gamma(a)).$$

So integrals over closed curves vanish. Conversely, on nice domains, vanishing around closed curves is closely related to the existence of a primitive. This is the beginning of Cauchy's theorem.

§1.10 Why singularities matter

The function $1/z$ has no primitive on $\mathbb{C} \setminus \{0\}$. Around the unit circle,

$$\int_{|z|=1} \frac{1}{z}\,dz = 2\pi i.$$

The nonzero integral detects the hole at the origin. This is the geometric heart of residue theory.

§1.11 A bridge outward

The important insight is that complex integration is simultaneously analytic and topological. Analytic regularity says holomorphic functions have local antiderivative-like behavior. Topology says singularities and holes can obstruct cancellation. The integral of $1/z$ around the unit circle is the prototype: the local function looks harmless away from 0, but the loop winds around a missing point and records $2\pi i$.

§1.12 Contour integrals as integrals of one-forms

A complex contour integral

$$\int_\gamma f(z)\,dz$$

integrates the complex one-form $f(z)\,dz$ along a parametrized curve. Under $z = \gamma(t)$, it becomes

$$\int_a^b f(\gamma(t))\gamma'(t)\,dt.$$

This is exactly the pullback rule for one-forms.

§1.13 Cauchy theorem and exactness

If f has a primitive F, then

$$\int_\gamma f(z)\,dz = F(\gamma(b)) - F(\gamma(a)).$$

Closed-loop integrals vanish. Cauchy–Goursat says holomorphic functions have this local exactness behavior on suitable domains.

§1.14 Residues as obstruction

For $f(z) = 1/z$, the integral around the unit circle is

$$\int_{|z|=1} \frac{dz}{z} = 2\pi i.$$

There is no global primitive on $\mathbb{C} \setminus \{0\}$. The residue measures the obstruction created by the puncture.

§1.15 Residues, topology, and exactness

Complex integration differs from real integration because holomorphicity, topology, and orientation interact. Contour integrals are one-form integrals; Cauchy theory is exactness; residues measure the failure of exactness around singularities.